9. Some money are being divided among Labib, Sami and Siam in the ratio of 5 : 4 : 2. If Siam gets Tk. 180, determine how much will Labib and Sami get ?

10. Sabuj, Dalim and Linkon are three brothers. Their father divided Tk. 6300 amongst them. Thus Sabuj gets $\frac{3}{5}$ parts of Dalim and Dalim gets double of Linkon. Find how much each of them will get.

11. A piece of ornament is made by mixing bronze, zinc and silver. In that piece of ornament, the ratio of bronze and zinc is 1 : 2 and the ratio of Zinc and silver is 3 : 5. Find how many grams of silver there are in an ornament weighing 19 grams.

12. Two equal size glasses are filled with sweet drink. In that sweet drink, the ratio of water and syrup in the first glass is 3 : 2 and in the second glass it is 5 : 4 respectively. If the sweet drink of two glasses are mixed together, find the ratio of water and syrup.

13. If $a : b = 4 : 7$, $b : c = 10 : 7$, find $a : b : c$.

14. If Tk. 9600 is divided amongst Sara, Mimuna and Raisa in the ratio of 4 : 3 : 1, how much taka will each of them get?

15. Tk. 4200 is divided amongst three students in the ratio of their classes. If they are the students of Class VI, VII and VIII, how many taka will each of them get ?

16. The ratio of income of Solaiman and Salman is 5 : 7. The ratio of income of Salman and Yousuf is 4 : 5. If the income of Solaiman is Tk. 120, what is the income of Yousuf ?

2·3 Profit-Loss

A shopkeeper bought one dozen ball pen at Tk. 60 and sold at Tk. 72. Here the shopkeeper bought 12 ball pen at Tk. 60. As a result the cost price of 1 ball pen is Tk. $\frac{60}{12}$ or Tk. 5. Again, he sold 12 ball pens at Tk. 72. As a result the selling price of 1 ball pen is Tk. $\frac{72}{12}$ or Tk. 6.

The cost price of 1 ball pen is Tk. 5. The selling price of 1 ball pen is Tk. 6.

The purchasing price of any thing is called **cost price** and the selling price of that thing is called **selling price.**

If the selling price is more than the cost price, then it is profitable.

$\therefore$ Profit = Selling price – Cost price

= Tk. 6 – Tk. 5 = Tk. 1.

Here, the shopkeeper gains a profit Tk. 1 for each ballpen.

Again, Suppose, a banana seller bought a bunch of four bananas at Tk. 20 and sold at Tk. 18. If the selling price is less than the cost price, then there is a loss.

$\therefore$ Loss = Cost price – Selling price

= Tk. (20–18) = Tk. 2.

Here the banana seller has a loss of Tk. 2 for each bunch of four bananas.

Suppose that, a cloth merchant who rented a shop appointed 5 employees. He bears, the rent of shop, employees' salaries, electric bill of the shop and miscellaneous expenses. All these costs are added to the cost price of the cloth. The sum of these is called total expenditure. If that merchant investing Tk. 2,00,000 sold cloth at Tk. 2,50,000 in a month, then he gained (Tk. 2,50,000 – Tk. 2,00,000) = Tk. 50,000 as profit. Again, if he sold cloth at Tk. 1,80,000, then he would have a loss of Tk. (2,00,000 – 1,80,000) = Tk. 20,000.

Observe :

- Profit = Selling price – Cost price
  or, Selling price = Cost price + Profit
  or, Cost price = Selling price – Profit
- Loss = Cost price – Selling price
  or, Cost price = Selling price + Loss
  or, Selling price = Cost price – Loss

We can express profit or loss in percentage. For example, in the discussion above we see that when a ballpen bought at Tk. 5 is sold at Tk. 6, there is a profit of Tk. 1.

That is, for Tk. 5 there is profit of Tk. 1

$\therefore$ „ Tk. 1 „ „ „ Tk. $\frac{1}{5}$

$\therefore$ „ Tk. 100 „ „ „ Tk. $\frac{1 \times \cancel{100}^{20}}{\cancel{5}_{1}} =$ Tk. 20

∴ The required profit is 20%.

The banana seller buying bananas at Tk. 20 sells at Tk. 18, then his loss is Tk. 2. That is,

for Tk. 20 there is loss of Tk. 2

∴ „ Tk. 1 „ „ „ Tk. $\frac{2}{20}$

∴ „ Tk. 100 „ „ „ Tk. $\frac{2 \times \cancel{100}^{5}}{\cancel{20}_{1}} = \text{Tk. } 10$

∴ The required loss is 10%.

Example 9. An orange seller bought 100 oranges for Tk. 1000 and sold them for Tk. 1200. What is his profit ?

Solution : The cost price of 100 oranges is Tk. 1000

The selling price of 100 oranges is Tk. 1200

Here, the selling price being more than cost price, it is profitable.

That is, Profit = selling price – cost price

= Tk. 1200 – Tk. 1000

= Tk. 200.

∴ The required profit is Tk. 200.

Example 10. A shopkeeper bought one sack of 50 kg of rice at Tk. 1600. Due to reduction of the price of rice, he sold it at Tk. 1500. What is his loss ?

Solution : Here,

the cost price of one sack of rice is Tk. 1600

and the selling price of one sack of rice is Tk. 1500.

∴ the selling price being less than the cost price, the shopkeeper bears a loss.

∴ Loss = cost price – selling price

= Tk. 1600 – Tk. 1500

= Tk. 100

∴ The required loss is Tk. 100

Example 11. If 15 ball pens are bought at Tk. 75 and are sold at Tk. 90, what is the percentage of profit ?

Solution : Here,
the cost price of 15 ball pens is Tk. 75
and the selling price of 15 ball pens is Tk. 90
the selling price being more than the cost price, it is profitable.

$\therefore$ Profit = selling price – cost price
= Tk. 90 – Tk. 75
= Tk. 15

$\therefore$ In Tk. 75 the profit is Tk. 15

„ Tk. 1 „ „ Tk. $\frac{15}{75}$

$\therefore$ „ Tk. 100 „ „ „ Tk. $\frac{\overset{1}{\cancel{15}} \times \overset{20}{\cancel{100}}}{\underset{1}{\cancel{75}_{\cancel{5}}}} = \text{Tk. } 20$

$\therefore$ The required profit is 20%.

Example 12. A fish seller bought four hilsha fishes at Tk. 1600 and sold each hilsha at Tk. 350. What is the percentage of his profit or loss ?

Solution : The price of 4 hilsha = Tk. 1600

$\therefore$ The price of 1 „ = Tk. $\frac{\overset{400}{\cancel{1600}}}{\cancel{4}}$

= Tk. 400

Again, the selling price of 1 hilsha is Tk. 350

Here, the selling price being less than the cost price, there is a loss.

$\therefore$ Loss = cost price – selling price
= Tk. 400 – Tk. 350
= Tk. 50

$\therefore$ For Tk. 400 there is loss of Tk. 50

„ Tk. 1 „ „ „ Tk. $\frac{50}{400}$

$\therefore$ „ Tk. 100 „ „ „ Tk. $\frac{\cancel{50}\times\cancel{100}^{1}}{\cancel{400}_{\cancel{4}_{2}}} = \text{Tk. } \frac{25}{2} = \text{Tk. } 12\frac{1}{2}$

$\therefore$ Loss is $12\frac{1}{2}\%$

Example 13. When a person sells a box of grapes for Tk. 2750, there is a loss of Tk. 450. If it is sold for Tk. 3600, then what is the profit or loss ?

Solution :

	Selling price of grapes	= Tk.	2750	
	loss	= Tk.	450	(Adding)
	cost price	= Tk.	3200	
Again,	selling price	= Tk.	3600	
	cost price	= Tk.	3200	(Subtracting)
	profit	= Tk.	400	

$\therefore$ Profit is Tk. 400.

Example 14. A tea seller bought one box of tea-leaves at the rate of Tk. 80 per kg. Selling the whole tea-leaves at the rate Tk. 75 per kg, he lost Tk. 500. How many kgs of tea-leaves did he buy ?

Solution : Cost price of per kg of tea-leaves = Tk. 80

Selling price of per kg of tea-leaves = Tk. 75

$\therefore$ Selling 1 kg of tea-leaves there, is a loss of Tk. 5

There is loss of Tk. 5 for 1 kg

„ „ „ Tk. 1 „ $\frac{1}{5}$ „

„ „ „ Tk. 500 „ $\frac{1\times\cancel{500}^{100}}{\cancel{5}_{1}}$ „

= 100 kgs

$\therefore$ 100 kgs tea-leaves was bought.

Example 15. An egg seller buys 5 dozen of eggs at the rate of Tk. 101 per dozen and 6 dozen at the rate of Tk. 90 per dozen. What should be the selling price if he wants to gain Tk. 3 per dozen?

Solution : Cost price of 1 dozen of eggs is Tk. 101

$\therefore$ „ „ „ 5 „ „ Tk. 101×5 = Tk. 505

Again, Cost price of 1 dozen of eggs is Tk. 90

„ „ „ 6 „ „ Tk. 90×6 = Tk. 540

$\therefore$ The cost price of (5+6) dozen or 11 dozen of eggs is Tk. (505 + 540) = Tk. 1045

$\therefore$ The cost price of 1 dozen of eggs = Tk. $\frac{1045}{11}$ = Tk. 95

On an average the cost price of 1 dozen of eggs is Tk. 95

For profit of Tk. 3 for each dozen of eggs, the selling price = Tk. (95 + 3) = Tk. 98

$\therefore$ If one dozen is sold at Tk. 98, the egg seller gets Tk. 3 as profit.

Example 16. A goat is sold at a loss of 10%. If the selling price is Tk. 450 more, then it is 5% profitable. What is the cost price ?

Solution : Let the cost price of the goat be Tk. 100.

In 10% loss the selling price is Tk. (100 – 10) = Tk. 90

In 5% profit the selling price Tk. (100 + 5) = Tk. 105

The selling price is in 5% profit – the selling price in 10% loss

= Tk. (105 – 90) or, Tk. 15

If the selling price is Tk. 15 more, then the cost price is Tk. 100

$\therefore$ „ „ „ „ „ Tk. 1 „ „ „ „ „ Tk. $\frac{100}{15}$

$\therefore$ „ „ „ „ „ Tk. 450 „ „ „ „ „ Tk. $\frac{100 \times \cancel{450}^{30}}{\cancel{15}_{1}}$

= Tk. 3,000

The cost price of the goat is Tk. 3000.

Example 17. Nabil bought 2 kgs of Sandesh at the rate of Tk. 250 per kg from a sweetshop. If the rate of VAT is Tk. 4 per 100. Tk., how many taka did he pay to the shopkeeper ?

Solution : The price of 1 kg Sandesh is Tk. 250

$\therefore$ „ „ „ 2 kg „ „ Tk. (250×2)

$= \text{Tk. } 500$

VAT for Tk. 100 is Tk. 4

$\therefore$ „ „ Tk. 1 „ Tk. $\frac{4}{100}$

$\therefore$ „ „ Tk. 500 „ Tk. $\frac{4 \times \cancel{500}^{5}}{\cancel{100}_{1}} = \text{Tk. } 20$

$\therefore$ Nabil paid Tk. $(500+20) = \text{Tk. } 520$ to shopkeeper.

Observation : Tax payable at a fixed rate with the cost price of a thing is called VAT (Value Added Tax).

Activity :

1. Kona bought a silk saree at Tk. 1,200 and a three piece at Tk. 1,800. If the rate of VAT is Tk. 4 percent, how many taka will she pay to the shopkeeper ?
2. Ishraq bought one dozen pencil at Tk. 250 from a stationery shop. If the rate of VAT is Tk. 4 percent, what is the price of each pencil ?

Example 18. Monthly basic pay of Mr. Nasir is Tk. 27,650. The tax for annual income for Tk. 2,50,000 as a first slab is Tk. 0. If the tax for next slab of income is at the rate of Tk. 10 per 100, how many taka does Mr. Nasir pay as income tax?

Solution : Monthly basic pay per month is Tk. 27,650

$\therefore$ „ „ „ „ for 12 months $= \text{Tk. } (27{,}650 \times 12)$

$= \text{Tk. } 3{,}31{,}800$

∴ The annual income tax payable for Tk. (3,31,800 – 2,50,000), or Tk. 81,800

The tax for Tk. 100 is Tk. 10

∴ „ „ „ Tk. 1 „ Tk. $\frac{10}{100}$

∴ „ „ „ Tk. 81,800 „ Tk. $\frac{10 \times \overset{818}{\cancel{81,800}}}{\underset{1}{\cancel{100}}}$ = Tk. 8180

∴ Mr. Nasir pays Tk. 8,180 as tax.

Example 19. If 1 US Dollar = Tk. 81·50 how much Bangladeshi currency does he need will be equal to 7,000 US Dollar?

Solution : 1 US Dollar = Tk. 81·50

7000 „ „ Tk. 81·50 × 7000

= Tk. 5,70,500·00

∴ The required amount of Bangladeshi currency is Tk. 5,70,500.

Exercise 2·2

1. If a shopkeeper bought 5 metres of cloth at the rate of Tk. 200 per metre and sold it at the rate of Tk. 225 per metre, the how much did he gain as profit ?
2. If an orange seller bought 5 dozen oranges at the rate of Tk. 60 per four and sold them at the rate of Tk. 50 per four, then how much did he lose ?
3. Rabi bought 50 kg rice at the rate of Tk. 40 per kg and sold it at the rate of Tk. 44 per kg. What is the amount of profit or loss ?
4. The buying rate of Milk vita milk Tk. 52 per litre. If it is sold at the rate of Tk. 55 per litre, then what is the percentage of profit ?